HR Attrition Analytics Dashboard

A Power BI dashboard that uses built-in AI visuals to identify what's actually driving employee attrition, paired with an interactive what-if simulator for testing retention interventions like overtime reduction and pay raises.

Overview

This project analyzes a synthetic HR dataset of 1,500 employees to answer two questions:

- What's actually causing people to leave?

using Power BI's Key Influencers AI visual and a Decompose Tree to surface statistically significant attrition drivers, rather than relying on assumptions.

- What happens if we intervene?

using What-If parameters to simulate the projected impact of reducing overtime or increasing pay, so HR can see estimated outcomes before committing budget to an intervention.

Key Findings

Based on the AI-driven Key Influencers analysis, the top predictors of attrition in this dataset are:

Driver	Impact
Overtime = Yes	1.94x more likely to leave
Low job satisfaction (score of 1)	1.89x more likely to leave
Poor work-life balance (score of 1)	1.54x more likely to leave
Distance from home > 20	1.42x more likely to leave
Age > 43	1.38x more likely to leave
Years since last promotion > 3	1.37x more likely to leave

Note, compensation (monthly income, salary hike %) did not rank among the top influencers — overtime and satisfaction-related factors matter more in this dataset than pay alone.

Dashboard Pages

- Overview — high-level KPIs (attrition rate, headcount) and page navigation
- Drivers — exploratory breakdown of attrition rate by department, overtime, job satisfaction, and work-life balance
- AI Insights — Power BI's Key Influencers and Decompose Tree AI visuals

- What-If Simulator — adjustable sliders for overtime reduction % and salary increase %, showing projected attrition rate impact in real time
- Employee Details (hidden, drill-through only) — individual employee records, reached by right-clicking any chart and selecting Drill through

Dataset

HR_Attrition_Dataset.xlsx — a synthetic dataset of 1,500 employee records (~19.7% attrition rate) with built-in, realistic correlations between attrition and factors like overtime, satisfaction scores, tenure, and pay. Generated for this project; not real employee data.

Columns include: demographics (Age, Gender, MaritalStatus), job info (Department, JobRole, TenureYears, YearsSinceLastPromotion), compensation (MonthlyIncome, PercentSalaryHike, StockOptionLevel), satisfaction scores (JobSatisfaction, EnvironmentSatisfaction, WorkLifeBalance, RelationshipSatisfaction, JobInvolvement), and work conditions (OverTime, BusinessTravel, DistanceFromHome).

How the What-If Simulator Works

Two What-If parameters drive a DAX measure (Projected Attrition Rate) that adjusts the baseline attrition rate downward based on slider position:

```
Projected Attrition Rate =
VAR BaseRate = [Attrition Rate]
VAR OvertimeSelected = [OverTimeReduction Value]
VAR SalarySelected = [SalaryIncrease Value]
VAR OvertimeEffect = (OvertimeSelected / 100) * 0.06
VAR SalaryEffect = (SalarySelected / 100) * 0.015
VAR Adjusted = BaseRate - OvertimeEffect - SalaryEffect
RETURN
MAX(Adjusted, BaseRate * 0.3)
```

Note: the coefficients (0.06, 0.015) are illustrative weights informed by the relative strength of each factor in the Key Influencers output — they are not derived from a statistical regression. For a production-grade version, these should be replaced with actual coefficients from a logistic regression model trained on the data.

Tools Used

- Power BI Desktop
- Power BI AI visuals: Key Influencers, Decompose Tree
- DAX (What-If parameters, calculated measures)

Setup

Clone this repository

Open HR_Attrition_Dashboard.pbix in Power BI Desktop

If prompted, update the data source path to point to HR_Attrition_Dataset.xlsx in your local copy of this repo

Refresh the data model

Project Status

Built as a learning/portfolio project to practice Power BI's AI-assisted analytics tools and DAX-driven what-if modeling.

License

MIT — feel free to fork and adapt for your own HR analytics practice.